

MATH 3330 Homework 3

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Question 1

Find if possible a choice function for the following collections, without using the choice axiom:

1. The collection $\mathcal{A}$ of nonempty subsets of $\mathbb{Z}_+$.
2. The collection $\mathcal{B}$ of nonempty subsets of $\mathbb{Q}$.

1. For a collection $\mathcal{A}$ of nonempty subsets of $\mathbb{Z}_+$, we define a choice function $f : \mathcal{A} \rightarrow \mathbb{Z}_+$ by $f(A) = \min(A)$ for all $A \in \mathcal{A}$. This is a valid choice function because every $A \in \mathcal{A}$ is nonempty, each $f(A)$ has a well defined smallest element, and $f(A) \in A$.
2. For a collection $\mathcal{B}$ of nonempty subsets of $\mathbb{Q}$, we define a choice function $f : \mathcal{B} \rightarrow \mathbb{Q}$. We will assume that all $\frac{p}{q} \in \mathbb{Q}$ is already in reduced form ($p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$). First we will define a well ordering " $\prec$ " on $\mathbb{Q}$:

$$\frac{p}{q} \prec \frac{p'}{q'} \iff \begin{cases} |p| + q < |p'| + q' & \text{or} \\ |p| + q = |p'| + q' \text{ and } q < q' & \text{or} \\ |p| + q = |p'| + q' \text{ and } q = q' \text{ and } p < p' \end{cases}$$

Now we define $f : \mathcal{B} \rightarrow \mathbb{Q}$ as $f(B) =$ "The smallest element in B according to $\prec$ " for all $B \in \mathcal{B}$. This is a valid choice function because every $B \in \mathcal{B}$ is nonempty, each $f(B)$ has a well defined smallest element and $f(B) \in B$.

Question 2

There was a theorem in §7 whose proof involved an infinite number of arbitrary choices. Which one was it? Rewrite the proof making explicit use of the choice axiom.

Theorem 7.5 in §7 uses infinitely many arbitrarily choices.

Proof: Let $(A_n)_{n \in \mathbb{N}}$ be a family of sets such that each A_n is countable. For each $n \in \mathbb{N}$ define

$S_n := \{f \in A_n^{\mathbb{N}} : f \text{ is surjective}\}$. Since A_n is countable, there exists at least one surjection $\mathbb{N} \rightarrow A_n$, so $S_n \neq \emptyset$ for every $n \in \mathbb{N}$.

By the Axiom of Choice applied to the indexed family $(S_n)_{n \in \mathbb{N}}$ of nonempty sets, there exists some choice function c on $\{S_n : n \in \mathbb{N}\}$. Let $A := \bigcup_{n \in \mathbb{N}} A_n$. We define $F : \mathbb{N} \times \mathbb{N} \rightarrow A$

as $F(n, m) = f_n(m)$. This is well defined since $f_n(m) \in A_n \subseteq A$ for all $(n, m) \in \mathbb{N} \times \mathbb{N}$.

We claim that F is surjective. Let $a \in A$. Then $a \in A_k$ for some $k \in \mathbb{N}$. Because f_k is surjective onto A_k , there exists $m \in \mathbb{N}$ such that $f_k(m) = a$. Therefore $F(k, m) = a$, so F is surjective.

Since $\mathbb{N} \times \mathbb{N}$ is countable, there exists a bijection $p : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$. Define $G : \mathbb{N} \rightarrow A$ by $G := F \circ p$. Since p is surjective and F is surjective, G is surjective.

Therefore there exists a surjection $\mathbb{N} \rightarrow A$. Hence A is countable.

Question 3

Let A and B be two nonempty sets. If there is an injection of B into A , but no injection of A into B , we say that A has greater cardinality than B . Show that if A has greater cardinality than B and B has greater cardinality than C , then A has greater cardinality than C .

Proof: Let A, B, C be nonempty sets. Let A have a greater cardinality than B , meaning that there exists an injection $f : B \rightarrow A$, but there does not exist an injection $f : A \rightarrow B$. Assume that A has a greater cardinality than C . Here we will denote a set having a greater cardinality than another set using the ">" symbol. Since $B > C$, there is an injection $g : C \rightarrow B$ and since $A > B$, there is an injection $f : B \rightarrow A$.

Now consider the composition of f and g , being $f \circ g : C \rightarrow A$. Because the composition of injective functions is injective, $f \circ g$ must be injective.

Then, suppose for the sake of contradiction that there exists an injection $h : A \rightarrow C$. Since g is injective, the composition $g \circ h : A \rightarrow B$ must also be injective by composition of injective functions. This contradicts the fact that $A > B$, so there must exist no injection h .

Therefore, A has a greater cardinality than C .

Question 4

Describe a well ordering on $\mathbb{Q}_+$ (Hint: use a well ordering on $\mathbb{Z}_+ \times \mathbb{Z}_+$). Using your ordering, find the smallest element in $(0, 1)$.

For $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$, we define the well ordering on $\mathbb{Q}_+$ as:

$$\frac{p}{q} < \frac{p'}{q'} \iff \begin{cases} |p| + q < |p'| + q' & \text{or} \\ |p| + q = |p'| + q' \text{ and } q < q' & \text{or} \\ |p| + q = |p'| + q' \text{ and } q = q' \text{ and } p < p' \end{cases}$$

for all $\frac{p}{q}, \frac{p'}{q'} \in \mathbb{Q}_+$.

The smallest element in $(0, 1)$ under this ordering would be $\frac{1}{2}$.